AI-driven interview and technical assessment platform

Project Team

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Chapter 1

Introduction

The increasing demand for technical professionals and the widespread adoption of remote hiring have transformed traditional interview and assessment processes. Modern technical interviews require evaluation across multiple dimensions, including programming skills, problem-solving ability, communication, and professional confidence. However, many existing assessment methods rely heavily on human judgment or isolated automated tools, resulting in inconsistent and incomplete evaluations.

Advancements in Artificial Intelligence, Natural Language Processing, and Computer Vision provide opportunities to enhance interview processes through automation and structured analysis. By integrating resume-based questioning, technical assessment, and behavioral observation, AI-assisted systems can support standardized and scalable evaluations. This project aims to leverage these technologies to create an intelligent mock interview platform that assists candidates and organizations by providing transparent and meaningful assessment outcomes.

1.1 Problem Statement

Traditional technical interviews and hiring assessments suffer from several limitations, including subjectivity, inconsistency, high operational cost, and limited scalability. Interview outcomes often depend on the interviewer's experience, mood, and bias, leading to unfair evaluations. Additionally, organizations rely on multiple disconnected platforms for resume screening, interviews, and coding assessments, resulting in fragmented candidate evaluation.

Furthermore, candidates rarely receive structured feedback, making it difficult for them to understand their weaknesses or improve their skills. Existing automated interview tools are either expensive, opaque in their scoring mechanisms, or focus on only one aspect of assessment such as coding or video interviews.

There is a need for an **integrated, transparent, and adaptive system** that can assess a candidate's technical knowledge, coding ability, communication skills, and behavioral indicators in a unified and objective manner.

1.2 Motivation

With the increasing demand for remote hiring and technical roles, organizations and educational institutions require scalable and cost-effective assessment solutions. Students and job seekers also need realistic mock interview platforms that simulate real interview environments and provide actionable feedback.

Motivated by the limitations of existing systems, this project aims to bridge the gap between **technical evaluation and behavioral analysis**, while maintaining transparency and explainability in AI-based decision making. The system is designed not to replace human interviewers, but to **assist and standardize** the interview and evaluation process.

1.3 Proposed Solution

This project proposes an AI-based Mock Interview and Technical Assessment System designed to provide end-to-end candidate evaluation through an integrated and structured framework. The system allows candidates to configure interviews based on their selected job role and difficulty level while also incorporating information extracted from the uploaded CV to generate relevant and personalized interview questions. The interview interaction is conducted in real time using both voice and text interfaces, simulating a realistic interview environment. Alongside the interview, an integrated online coding environment enables candidates to solve technical problems, allowing the system to evaluate both conceptual understanding and practical coding ability.

The proposed solution employs automated evaluation techniques to assess answer relevance using Natural Language Processing, analyze code correctness and execution results, and observe behavioral indicators such as eye movement, speech patterns, and facial cues using Computer Vision and audio analysis. These multiple evaluation dimensions are combined using a transparent scoring mechanism that integrates technical performance, communication skills, and behavioral indicators. The system generates a comprehensive feedback report highlighting candidate strengths, weaknesses, and areas for improvement.

1.4 Key Benefits

The proposed system offers a holistic evaluation of candidates that goes beyond traditional interview approaches by integrating technical, behavioral, and communication assessments into a single platform. It reduces interviewer bias by applying standardized and objective assessment criteria across all candidates. The system also provides personalized and constructive feedback, enabling candidates to better understand their performance and focus on skill improvement. Additionally, it supports scalable mock interview and hiring simulations, making it suitable for educational institutions, training centers, and organizations. The platform also serves as a

valuable tool for academic research in AI-based assessment and explainable artificial intelligence.

1.5 Operational Efficiency

The proposed system significantly improves operational efficiency by automating repetitive interview and screening tasks that traditionally require extensive human effort. By reducing the dependence on human interviewers, the system minimizes workload, scheduling conflicts, and manual evaluation delays. It enables multiple candidates to be assessed simultaneously without compromising evaluation consistency. Furthermore, instant generation of evaluation reports eliminates the need for post-interview analysis, resulting in faster assessments, reduced operational costs, and consistent evaluation quality.

1.6 Stakeholders

The key stakeholders of the system include:

1. **Job Candidates / Students**
    - Receive realistic mock interviews
    - Gain structured feedback for skill improvement
2. **Recruiters / HR Professionals**
    - Obtain standardized and objective candidate evaluations
    - Reduce interview time and effort
3. **Educational Institutions**
    - Use the platform for interview preparation and skill assessment
    - Support student career development
4. **Researchers and Developers**
    - Analyze AI-based evaluation methods
    - Study bias, transparency, and assessment accuracy

Chapter 2

Project Description

2.1 Scope

The scope of this project encompasses the design, development, and evaluation of an AI-based mock interview and technical assessment platform aimed at supporting interview preparation, candidate evaluation, and academic experimentation. The system focuses on role-based interview configuration, where candidates can select predefined job roles and difficulty levels to simulate real-world interview scenarios. It includes automated resume parsing to extract relevant skills and experience, enabling personalized and context-aware interview question generation. The interview process is conducted through real-time voice and text-based interaction, providing a realistic and engaging environment for candidates. Additionally, the system integrates an online coding environment that allows candidates to solve technical problems during the interview, enabling simultaneous assessment of theoretical understanding and practical coding skills. The platform applies Natural Language Processing techniques to analyze answer relevance, evaluates code execution results for correctness, and observes behavioral indicators such as speech patterns, facial movement, and eye direction using computer vision and audio analysis techniques. The primary focus of the system is to deliver structured, transparent, and explainable evaluation outcomes rather than subjective judgments.

The scope of the project is intentionally limited to ensure feasibility within academic and time constraints. The system supports a restricted set of job roles, programming languages, and evaluation metrics, and the behavioral analysis is based on observable indicators rather than deep psychological or emotional inference. The platform is not intended to replace professional recruiters, conduct real hiring decisions, or guarantee employment outcomes. Instead, it serves as a decision-support and learning tool that assists candidates in understanding their performance and helps organizations or educational institutions conduct standardized assessments. Large-scale enterprise deployment, real-time bias mitigation, advanced emotion recognition, and legal compliance frameworks are considered outside the scope of this project.

2.2 Modules

2.2.1 User Authentication and Profile Management Module

This module manages user registration, login, and role selection within the system. It allows candidates to choose their desired job role and difficulty level and securely upload their CV. The module ensures authenticated access and maintains user profiles for personalized interview sessions.

2.2.2 Resume Parsing and Skill Extraction Module

This module processes uploaded CVs to extract relevant information such as skills, experience, projects, and tools using Natural Language Processing techniques. The extracted data is used to create a structured candidate profile. This enables personalized interview question generation based on the candidate's background.

2.2.3 Interview Configuration and Question Generation Module

This module generates interview questions based on the selected job role, difficulty level, and extracted resume content. It uses AI-driven logic to create technical and conceptual questions dynamically. The module ensures that interviews are tailored and relevant rather than generic.

2.2.4 AI Interview Interaction Module

This module conducts the mock interview through real-time voice and text-based interaction. It uses speech-to-text and text-to-speech technologies to enable natural communication between the candidate and the system. The module manages interview flow, question sequencing, and response capture.

2.2.5 Coding Assessment and Execution Module

This module provides an integrated online coding environment where candidates solve programming problems during the interview. The system executes the submitted code in a controlled environment and evaluates compilation success and output correctness. This enables practical assessment of programming skills.

2.2.6 Answer Analysis and Technical Evaluation Module

This module analyzes candidate responses using Natural Language Processing techniques to determine answer relevance and conceptual accuracy. It compares responses against expected technical concepts. The module generates a technical score based on predefined evaluation criteria.

2.2.7 Behavioral and Communication Analysis Module

This module evaluates observable behavioral indicators such as eye movement, facial orientation, speech patterns, and hesitation levels using computer vision and audio analysis. The analysis focuses on measurable indicators rather than emotional inference. The results contribute to the communication and behavioral score.

2.2.8 Scoring and Explainable Evaluation Module

This module combines technical, coding, communication, and behavioral scores into a final evaluation. It applies a transparent and explainable scoring mechanism where each metric contributes a defined weight. The module ensures fairness and avoids black-box decision making.

2.2.9 Feedback and Report Generation Module

This module generates a detailed evaluation report highlighting the candidate's strengths, weaknesses, and areas for improvement. The report provides actionable recommendations for skill enhancement. It serves as a learning and self-assessment tool for candidates.

2.2.10 Admin and System Management Module

This module allows administrators to manage job roles, question banks, difficulty levels, and evaluation parameters. It supports monitoring system performance and maintaining assessment standards. The module ensures controlled and consistent system operation.

2.3 Tools and Technologies

Programming Languages

- **Python –** Used for AI/ML model development, computer vision algorithms, and data analysis.
- **JavaScript –** Enables frontend development and client-side interactivity.
- **TypeScript –** Provides type safety and scalability, enhancing JavaScript for largescale applications.

Frontend Technologies

- **React.js –** Component-based framework for building dynamic and interactive user interfaces.
- **Next.js –** React framework supporting server-side rendering (SSR) and optimized web application performance.
- **Tailwind CSS –** Utility-first CSS framework for rapid UI prototyping and responsive designs

Backend Technologies

- **FastAPI –** High-performance Python framework for building RESTful APIs with built-in validation.
- **Node.js –** JavaScript runtime for executing server-side logic and scalable backend services.
- **Express.js –** Web application framework for Node.js, providing routing and middleware functionalities.

Database Technologies

- **PostgreSQL –** Relational database management system for structured data and transactional integrity.
- **MongoDB –** NoSQL database for flexible, document-oriented storage of unstructured or semi-structured data.

2.4 Work Division

Name	Reg No	Work Assigned
Ahsan Iqbal	I222524	Behavioral analysis, Computer Vision, report generation
Zain ali Khan	I22	Backend development, AI & NLP integration, CV parsing
Khizar Shahid	I222595	Frontend development, UI/UX design, coding module

2.5 Timeline

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